Yaoxuan Zeng

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APPOINTMENTS

School of Science Dean's Postdoctoral Fellow, MIT — Cambridge, MA, USA
Supervised by Prof. Wanying Kang — Sep 2026 – present

Woods Hole Oceanographic Institution — Woods Hole, MA, USA
GFD fellow at the Geophysical Fluid Dynamics Program — Jun 2023 – Aug 2023

Peking University, School of Physics — Beijing, China
Research Assistant with Prof. Jun Yang — Sep 2020 – Jul 2021

EDUCATION

The University of Chicago, Department of the Geophysical Sciences — Chicago, IL, USA
Ph.D. in Geophysical Sciences (advisor: Malte F. Jansen) — Sep 2021 – Aug 2026
Thesis title: Ocean circulation on icy moons

Peking University, School of Physics — Beijing, China
B.S. in Atmospheric and Oceanic Sciences (Advisor: Jun Yang) — Sep 2016 – Jul 2020
Undergrad project: Oceanic superrotation on tidally locked planets

PUBLICATIONS

Published

1. Heng Quan, Da Yang, William Boos, Tiffany Shaw, Huazhi Ge, **Yaoxuan Zeng**, and Carly KleinStern: Water vapor buoyancy and the African easterly jet. *arXiv preprint arXiv:2605.21875* (2026).
2. **Yaoxuan Zeng**, Wanying Kang, Glenn Flierl, and Geoff Vallis: Influence of Penetration Depth on Jets on Giant Planets: Equatorial Jet Direction, Jet Numbers, and Jet Energy Fraction. *The Planetary Science Journal*, 7, 149 (2026).
3. **Yaoxuan Zeng**, and Malte F. Jansen: Slantwise convection and heat transport in icy moon oceans. *Geophysical Research Letters*, 53, e2025GL118761 (2026).
4. **Yaoxuan Zeng**, and Malte F. Jansen: Symmetric instability in a Boussinesq fluid on a rotating planet. *Journal of Geophysical Research: Planets*, 130, e2025JE009115 (2025).
5. **Yaoxuan Zeng**, and Malte F. Jansen: The effect of salinity on ocean circulation and ice-ocean interaction on Enceladus. *The Planetary Science Journal*, 5, 13 (2024).
6. Malte F. Jansen, Wanying Kang, Edwin Kite, and **Yaoxuan Zeng**: Energetic constraints on ocean circulation of icy ocean worlds. *The Planetary Science Journal*, 4, 117 (2023).
7. **Yaoxuan Zeng**, and Malte F. Jansen: Ocean Circulation on Enceladus with a High- versus Low-salinity Ocean. *The Planetary Science Journal*, 2, 151 (2021).
8. **Yaoxuan Zeng**, and Jun Yang: Oceanic Superrotation on Tidally Locked Planets. *The Astrophysical Journal*, 909, 172 (2021).
9. Jun Yang, Weiwen Ji and **Yaoxuan Zeng**: Transition from Eyeball to Snowball Driven by Sea-ice Drift on Tidally Locked Terrestrial Planets. *Nature Astronomy*, 4, 58–66 (2020).

Under review

10. **Yaoxuan Zeng**, and Malte F. Jansen: Parameterizing slantwise convection in icy moon oceans. *arXiv preprint arXiv:2606.22547* (2026).

Manuscript in preparation

11. **Yaoxuan Zeng**, and Malte F. Jansen: Interplay between slantwise convection and baroclinic eddies in the icy moon oceans. In preparation.

SELECTED PRESENTATIONS

1. Slantwise convection and heat transport in icy moon oceans. 25th Conference on Atmospheric and Oceanic Fluid Dynamics, Houston, TX, USA, Jan 2026.
2. Slantwise convection in icy moon oceans. 2024 AGU Annual Meeting, Washington DC, USA, Dec 2024.
3. Symmetric instability of Boussinesq fluid on a rotating planet. 24th Conference on Atmospheric and Oceanic Fluid Dynamics, Burlington, VT, USA, June 2024.
4. Ocean circulation on Enceladus: the effect of salinity on ocean heat and tracer transport. Purdue Storm Snack seminar (invited), West Lafayette, IN, USA, Jan 2024.
5. Jets on Giant Planets – A Tale of Two Forcings. GFD fellow project presentation, WHOI, Woods Hole, MA, USA, Aug 2023.
6. The effect of gas bubbles on nanoparticle transport and ocean circulation on Enceladus. 2022 AGU Fall Meeting (poster), Chicago, IL, USA, Dec 2022.
7. Ocean Circulation on Enceladus with A High- Versus Low-salinity Ocean. 23rd Conference on Atmospheric and Oceanic Fluid Dynamics, Breckenridge, CO, USA, June 2022.
8. Wind-driven Ocean Circulation on Tidally Locked Aqua-planets. Astrobiology Science Conference, Bellevue, WA, USA, Jun 2019.

SELECTED AWARDS AND HONORS

- **School of Science Postdoctoral Fellowship** at MIT 2026
- **Second Place Best AOFD Student Oral Presentation** of American Meteorological Society, 106th AMS Annual conference 2026
- **GFD fellowship** of WHOI 2023
- **First Place Outstanding Student Presentation Award: Oral** of American Meteorological Society, 23rd AOFD conference 2022
- **Neubauer Graduate Scholarship** of UChicago 2021-2026

TEACHING

- **Lecturer** for Global Warming: Understanding the Forecast 2021/22/24
- **Teaching Assistant** for Geophysical Fluid Dynamics: Foundations 2025
- **Teaching Assistant** for Geophysical Fluid Dynamics: Rotation and Stratification 2024/26

PROFESSIONAL SERVICE

- **Core organizer** for **Rossbypalooza Summer School** at UChicago 2024
- **Organizer** for GeoSci Grad Expo at UChicago 2024
- **Co-organizer** for Earth Science Day for High School Students at UChicago 2023
- **Organizer** for Climate Journal Club and Dynamics group meeting 2022-2026
- **Reviewer** for *Communications Earth & Environment*; *Icarus*